Classical Field Theory — Exercise Sheet 6

Causal kernels and frequency boundary values

Solutions

Use $\Box = \partial_t^2 - \Delta$ and Lecture 6's normalized kernel $G_{\mathrm{ret}} = \delta(t-|x|)/(4\pi|x|)$ in three spatial dimensions. Its meaning at the vertex is its test pairing, not a product of distributions. All distribution pairings are bilinear. Write H for the Heaviside function; its value at zero does not affect the distributions below.

Exercise 6.1 (An entire radial source response). Let $q \in C_c^\infty(\mathbb{R})$, and let h be a smooth even compactly supported function such that $x \mapsto h(|x|)$ is smooth on $\mathbb{R}^3$. Take $f(t,x) = q(t)h(|x|)$ and $u = G_{\mathrm{ret}} * f$.

(a) Perform the angular integration to obtain a one-dimensional radial integral for $u(t,r)$, $r = |x| > 0$, with an inner integral over travel distance. Find its limit at $r = 0$ and explain why no singularity remains there.

(b) Set $v(t,r) = ru(t,r)$ for $r \geq 0$ and extend it oddly to the real line. Reduce the equation for v to a forced one-dimensional wave and derive its causal double-integral formula. Check its agreement with part (a).

(c) Let $a < b$ and $R > 0$. If $\operatorname{supp} q \subset [a,b]$ and $\operatorname{supp} h \subset [-R,R]$, find the possible response interval at an observer $r > R$. If $q > 0$ on (a,b) and $h > 0$ for $|s| < R$, prove that the response is positive throughout the interior of that interval.

Solution. Lecture 6 gives a smooth rotation-invariant response by integrating $q(t-|x-z|)h(|z|)/(4\pi|x-z|)$ over z. For $r,s > 0$, put $\mu = \cos\theta$. Then $\ell^2 = r^2 + s^2 - 2rs\mu$ and $\mathrm{d}\mu = -\ell\,\mathrm{d}\ell/(rs)$. The azimuthal integral is 2π. Consequently

$$u(t,r) = \frac{1}{2r}\int_0^\infty sh(s)\int_{|r-s|}^{r+s} q(t-\ell)\,\mathrm{d}\ell\,\mathrm{d}s.$$

For fixed $s > 0$, the inner integral divided by $2r$ tends to $q(t-s)$ as $r \downarrow 0$. Its absolute value is at most $\|q\|_\infty$, since its interval has length $2\min(r,s) \leq 2r$. Dominated convergence with the integrable majorant $s|h(s)|\|q\|_\infty$ proves

$$u(t,0) = \int_0^\infty sh(s)q(t-s)\,\mathrm{d}s.$$

This also follows directly from the original three-dimensional integral at $x = 0$. Smoothness at the centre is supplied by Lecture 6's convolution theorem, not merely by finiteness of this radial limit. Radial smoothness implies $u_r(t,0) = 0$.

For the second part, the radial Laplacian identity gives $v_{tt} - v_{rr} = rq(t)h(r)$ for $r > 0$. The smooth even radial profile of u makes v a smooth odd function of r, and the odd forcing is $rq(t)h(|r|)$. The causal one-dimensional formula is

$$v(t,r) = \frac{1}{2}\int_{-\infty}^{t} q(\tau)\int_{r-(t-\tau)}^{r+(t-\tau)} sh(|s|)\,\mathrm{d}s\,\mathrm{d}\tau.$$

To verify it without assuming a Duhamel theorem, differentiate the inner endpoint integral twice. Its zero-length interval at $\tau = t$ contributes $rh(|r|)q(t)$ to v_{tt}; the remaining terms equal v_{rr}. All integrals and derivatives are justified on compact sets by compact support

and smoothness. The formula vanishes before q starts, so Lecture 5's one-dimensional local uniqueness identifies it with ru.

There is also a direct equality of the two integrals. Splitting the odd source into positive and negative s and interchanging integrals yields, for $r \geq 0$,

$$v(t,r) = \frac{1}{2}\int_0^\infty sh(s)\left[\int_{-\infty}^{t-|r-s|} q(\tau)\mathrm{d}\tau - \int_{-\infty}^{t-r-s} q(\tau)\mathrm{d}\tau\right]\mathrm{d}s.$$

The bracket is $\int_{|r-s|}^{r+s} q(t-\ell)\mathrm{d}\ell$, proving agreement.

For the final part, a source position in $B_R(0)$ is at distance between $r-R$ and $r+R$ from an exterior observer. The retarded argument must be in $[a,b]$, so the field vanishes outside $[a+r-R, b+r+R]$. Under the positivity hypotheses, an interior time satisfies

$$(r-R, r+R) \cap (t-b, t-a) \neq \varnothing.$$

Choose a travel distance in this open intersection. The corresponding source points include an open set inside $B_R(0)$ on which both factors of the original integrand are positive. Its integral is positive. At the endpoints the field is zero by smoothness and its vanishing on the adjacent exterior time intervals. The strict positivity assertion requires these extra hypotheses; a support bound alone would allow cancellation.

Exercise 6.2 (Which side of the frequency poles?). The spacetime Fourier convention is

$$\widetilde{F}(\nu,k) = (2\pi)^{-2}\int e^{i\nu t - ik\cdot x}F(t,x)\mathrm{d}t\mathrm{d}^3x.$$

For a tempered distribution it is defined by transposition on Schwartz tests, with this same kernel. Fourier transformation preserves Schwartz space and its topology, so this definition preserves convergence of tempered distributions. The Schwartz Fourier automorphism is the one from *Analysis* I used in Lecture 4; transposition here specifies its distributional extension. Let $G_\varepsilon = e^{-\varepsilon t}G_{\mathrm{ret}}$, $\varepsilon > 0$; the product is understood on the future cone, where the factor is bounded.

(a) Compute its spatial transform, then its full spacetime transform. Prove convergence as $\varepsilon \downarrow 0$ on Schwartz tests and express $\widetilde{G}_{\mathrm{ret}}$ as a frequency boundary value. Check the Fourier coefficient of the equation $\Box G_{\mathrm{ret}} = \delta^4$.

(b) For fixed $\kappa = |k| > 0$, derive the principal-value and delta terms of that boundary value. Here $\langle \mathrm{pv}(1/s), \zeta\rangle = \lim_{a\downarrow 0}\int_{|s|>a}\zeta(s)/s\,\mathrm{d}s$. Derive the identity for $(s \pm i0)^{-1}$ by separating real and imaginary parts. Compare the advanced and half-retarded plus half-advanced kernels. State why the fixed-κ decomposition cannot simply be evaluated at $\kappa = 0$.

Solution. The spatial transform includes the factor $(2\pi)^{-3/2}$. Angular integration on the shell of radius t therefore gives

$$\widehat{G}_\varepsilon(t,k) = (2\pi)^{-3/2}H(t)e^{-\varepsilon t}\frac{\sin(t\kappa)}{\kappa}, \qquad \kappa = |k|,$$

with continuous value t for the quotient at $\kappa = 0$. Splitting the sine into exponentials and integrating on $t \geq 0$ gives

$$\int_0^\infty e^{-(\varepsilon - i\nu)t}\frac{\sin(t\kappa)}{\kappa}\mathrm{d}t = \frac{1}{(\varepsilon - i\nu)^2 + \kappa^2} = \frac{1}{\kappa^2 - (\nu + i\varepsilon)^2}.$$

The remaining unitary time factor is $(2\pi)^{-1/2}$. Thus the full transform is

$$\widetilde{G}_\varepsilon = (2\pi)^{-2}\frac{1}{|k|^2-(\nu+i\varepsilon)^2},$$

also at $\kappa = 0$ by its integral limit.

The cone pairing with a Schwartz test is bounded by a constant times $\int_0^\infty r(1+r)^{-N}\mathrm{d}r$ for $N > 2$. The same majorant works with $e^{-\varepsilon r} \le 1$, so dominated convergence gives $G_\varepsilon \to G_{\mathrm{ret}}$ as tempered distributions. Transposing the continuous Schwartz Fourier map passes this convergence to the transforms. Thus

$$\widetilde{G}_{\mathrm{ret}} = (2\pi)^{-2}\lim_{\varepsilon\downarrow 0}\frac{1}{|k|^2-(\nu+i\varepsilon)^2},$$

where the limit is a distribution on all four frequency variables. Transforming the already proved fundamental-solution equation gives $(|k|^2-\nu^2)\widetilde{G}_{\mathrm{ret}} = (2\pi)^{-2}$, since the transform of δ^4 is the constant $(2\pi)^{-2}$. It is not the constant one in the unitary convention.

For the second part, write

$$\frac{1}{s\pm i\varepsilon} = \frac{s}{s^2+\varepsilon^2} \mp i\frac{\varepsilon}{s^2+\varepsilon^2}.$$

For a Schwartz test, substitute $s = \varepsilon z$ in the second term. Dominated convergence gives its limit $\pi\zeta(0)$. For the first term subtract $\zeta(0)$ times an even cutoff equal to one near zero; the subtracted part has zero odd integral. The remaining test vanishes at zero, so dominated convergence near zero and at infinity gives the principal value. Consequently $(s\pm i0)^{-1} = \mathrm{pv}(1/s) \mp i\pi\delta(s)$. At a fixed $\kappa > 0$, the partial fractions are

$$\frac{1}{\kappa^2-(\nu+i\varepsilon)^2} = \frac{1}{2\kappa}\left[-\frac{1}{\nu-\kappa+i\varepsilon}+\frac{1}{\nu+\kappa+i\varepsilon}\right].$$

Applying the identity gives

$$\lim_{\varepsilon\downarrow 0}\frac{1}{\kappa^2-(\nu+i\varepsilon)^2} = \mathrm{pv}(\kappa^2-\nu^2)^{-1} + \frac{i\pi}{2\kappa}[\delta(\nu-\kappa)-\delta(\nu+\kappa)].$$

The principal value at the two distinct roots is understood by the same partial fractions. Time reflection gives the advanced denominator $\kappa^2-(\nu-i0)^2$. Its imaginary term has the opposite sign, so the half-sum cancels it. The factor $1/(2\kappa)$ and the two-root separation require $\kappa > 0$. The full boundary-value limit above already includes zero spatial frequency; one must not replace it by an undefined substitution into the separated terms.

Exercise 6.3 (Descending the spatial dimension). Starting with the three-dimensional retarded pairing, integrate out one auxiliary spatial coordinate to construct a kernel G_2 on $\mathbb{R}^{1+2}$, and two auxiliary coordinates to construct G_1 on $\mathbb{R}^{1+1}$. Make the construction legitimate on compact smooth tests: a test independent of the auxiliary coordinates is not itself compactly supported. Derive explicit locally integrable formulas, prove $(\partial_t^2-\Delta_d)G_d = \delta^{1+d}$ for $d = 1, 2$, and identify their support and their relation to Sheet 5's interior tails.

Solution. Let $\psi(t,x)$ be a compact test in the retained coordinates, with $|t| \le T$ on its support. Lift it to $\psi(t,x)\zeta(z)$, where the auxiliary cutoff is one for $|z| \le T+1$. On the three-dimensional cone $t = \sqrt{|x|^2+|z|^2}$, a nonzero lifted test necessarily has $|z| \le T$. The resulting pairing is independent of any larger cutoff and defines G_d. Its local continuity follows from the three-dimensional pairing with this fixed cutoff for tests of fixed support.

For one removed coordinate, the two roots of $t = \sqrt{r^2 + z^2}$, for $t > r = |x|$, are $z = \pm\sqrt{t^2 - r^2}$. Changing variables on the two half-lines gives

$$G_2(t, x) = \frac{H(t - r)}{2\pi\sqrt{t^2 - r^2}}.$$

This fraction is evaluated for $t > r$ and the function is zero outside that open cone; values on the cone itself do not affect its distribution. This is a locally integrable function: for $0 < t < T$, its spatial radial integral contains $\int_0^t r(t^2 - r^2)^{-1/2}\mathrm{d}r = t$. The change of variables in the original pairing also proves that no additional cone-supported atom was omitted.

For two removed coordinates, let their radius be s and set $\ell = \sqrt{x^2 + s^2}$. The polar Jacobian gives

$$\int_0^\infty \frac{2\pi s}{4\pi\sqrt{x^2 + s^2}}\delta(t - \sqrt{x^2 + s^2})\mathrm{d}s = \frac{1}{2}\int_{|x|}^\infty \delta(t - \ell)\mathrm{d}\ell = \frac{1}{2}H(t - |x|).$$

These delta expressions abbreviate changes of variables on tests, so there is no multiplication of singular distributions.

Finally apply the three-dimensional identity to the lifted test. The retained wave operator acts on ψ; the extra term differentiating ζ is supported where $|z| > T + 1$ and hence has zero cone pairing. The vertex has $\zeta(0) = 1$. Therefore

$$\langle G_d, (\partial_t^2 - \Delta_d)\psi\rangle = \psi(0, 0), \qquad d = 1, 2.$$

The formulas vanish outside the future cone and are strictly positive in its interior. Their supports are its whole closure. Their convolution with initial velocity gives exactly the one- and two-dimensional formulas used on Sheet 5, explaining why an interior tail is compatible with finite speed.